

Лабораторная работа № 3

Измерение и тестирование пропускной способности сети. Воспроизводимый эксперимент

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Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

1. Воспроизвести посредством API Mininet эксперименты по измерению пропускной способности с помощью iPerf3.
2. Построить графики по проведённому эксперименту.

Выполнение лабораторной работы

```
mininet-vn login: mininet
Password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-216-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
Last login: Fri Sep 26 05:14:55 PDT 2025 from 192.168.56.1 on pts/0
mininet@mininet-vm:~$
mininet@mininet-vm:~$ cd ~/work/lab_iperf3
mininet@mininet-vm:~/work/lab_iperf3$ mkdir lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/lab_iperf3_topo
-bash: cd: /home/mininet/work/lab_iperf3/lab_iperf3_topo: No such file or directory
mininet@mininet-vm:~/work/lab_iperf3$
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp ~/mininet/examples/emphynet.py ~/work/lab_iperf3/lab_iperf3_topo
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv emphynet.py lab_iperf3_topo.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ _
```

Figure 1: Копирование файла emphynet.py

Выполнение лабораторной работы

```
#!/usr/bin/env python
"""
This example shows how to create an empty Mininet object
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():
    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3 )
    net.addLink( h2, s3 )

    info( '*** Starting network\n' )
    net.start()

    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()
```

Figure 2: Содержание файла lab_iperf3_topo.py

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
o.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
*** Running CLI
*** Starting CLI:
mininet> net
*** Unknown command: tnet
mininet> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
s3 lo: s3-eth1:h1-eth0 s3-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s3-eth1 (OK OK)
h2-eth0<->s3-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=898>
<Host h2: h2-eth0:10.0.0.2 pid=901>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=906>
<Controller c0: 127.0.0.1:6653 pid=891>
```

Figure 3: Создание топологии и ее основные параметры

```
"Create an empty network and add nodes to it."

net = Mininet( controller=Controller, waitConnected=True )

info( '*** Adding controller\n' )
net.addController( 'c0' )

info( '*** Adding hosts\n' )
h1 = net.addHost( 'h1', ip='10.0.0.1' )
h2 = net.addHost( 'h2', ip='10.0.0.2' )

info( '*** Adding switch\n' )
s3 = net.addSwitch( 's3' )

info( '*** Creating links\n' )
net.addLink( h1, s3 )
net.addLink( h2, s3 )

info( '*** Starting network\n' )
net.start()
print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
print( "Host", h2.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )
info( '*** Running CLI\n' )
CLI( net )

info( '*** Stopping network' )
net.stop()
```

Figure 4: Изменение скрипта lab_iperf3_topo.py

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_i
o.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address a2:ec:e0:b8:20:47
Host h2 has IP address 10.0.0.2 and MAC address 0e:d1:bb:fc:ee:16
*** Running CLI
*** Starting CLI:
```

Figure 5: Проверка работы внесенных изменений

```
GNU nano 4.8 lab_iperf3_topo2.py
(without a topology object) and add nodes to it manually.
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.node import CPULimitedHost
from mininet.link import TCLink
from mininet.cli import CLI
from mininet.log import setLogLevel, info

def emptyNet():
    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True, host = CPULimitedHost, link = TCLink )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1', cpu=50 )
    h2 = net.addHost( 'h2', ip='10.0.0.2', cpu=45 )

    info( '*** Adding switch\n' )
    s3 = net.addSwitch( 's3' )

    info( '*** Creating links\n' )
    net.addLink( h1, s3, bw=10, delay='5ms',max_queue_size=1000, loss=10, use_htb=True )
    net.addLink( h2, s3 )

    info( '*** Starting network\n' )
    net.start()
    print( "Host", h1.name, "has IP address", h1.IP(), "and MAC address", h1.MAC() )
    print( "Host", h2.name, "has IP address", h2.IP(), "and MAC address", h2.MAC() )
    info( '*** Running CLI\n' )
    CLI( net )

    info( '*** Stopping network' )
    net.stop()
```

Figure 6: Настройка параметров производительности

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ sudo python lab_iperf3_topo.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s3 ...
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC address 66:23:c2:cf:fd:9b
Host h2 has IP address 10.0.0.2 and MAC address 12:8a:76:57:97:a7
*** Running CLI
*** Starting CLI:
```

Figure 7: Запуск скрипта с настройкой параметров производительности и без нее

```
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cp lab_iperf3_topo2.py lab_iperf3.py
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mkdir -p ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ mv ~/work/lab_iperf3/lab_iperf3_topo/lab_iperf3.py ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/lab_iperf3_topo$ cd ~/work/lab_iperf3/iperf3
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ ls -l
total 4
-rwxrwxr-x 1 mininet mininet 1341 Oct  7 14:02 lab_iperf3.py
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

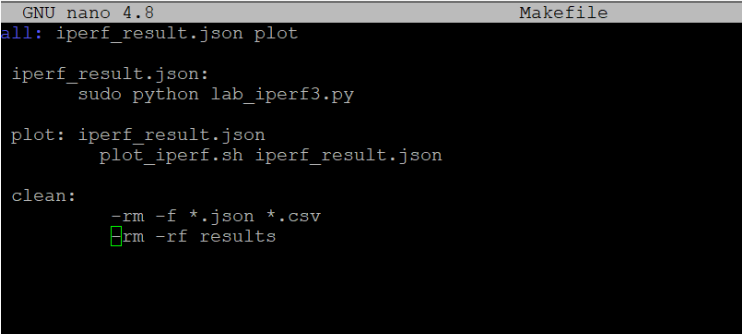
Figure 8: Создание копии скрипта lab_iperf3_topo2.py

```
def emptyNet():  
    "Create an empty network and add nodes to it."  
  
    net = Mininet( controller=Controller, waitConnected=True, host = CPULimitedHost, link = TCLink )  
  
    info( '*** Adding controller\n' )  
    net.addController( 'c0' )  
  
    info( '*** Adding hosts\n' )  
    h1 = net.addHost( 'h1', ip='10.0.0.1' )  
    h2 = net.addHost( 'h2', ip='10.0.0.2' )  
  
    info( '*** Adding switch\n' )  
    s3 = net.addSwitch( 's3' )  
  
    info( '*** Creating links\n' )  
    net.addLink( h1, s3, bw=100, delay='75ms' )  
    net.addLink( h2, s3, bw=100, delay='75ms' )  
  
    info( '*** Starting network\n' )  
    net.start()  
    info( '*** Starting network\n' )  
  
    info( '*** Traffic generation\n' )  
    h2.cmdPrint( 'iperf3 -s -D -l' )  
    time.sleep(10) # Wait 10 seconds for servers to start  
    h1.cmdPrint( 'iperf3 -c', h2.IP(), '-J > iperf_result.json' )  
  
    # info( '*** Running CLI\n' )  
    # CLI( net )  
  
    # info( '*** Running CLI\n' )  
    # CLI( net )  
  
    info( '*** Stopping network' )  
    net.stop()
```

Figure 9: Изменен ия кода в скрипте lab_iperf3.py

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(10.00Mbit 5ms delay 10.00000% loss) (10.00Mbit 5ms delay 10.00000% loss) *** Starting network
*** Configuring hosts
h1 (cfs 20000000/100000us) h2 (cfs 18000000/100000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (10.00Mbit 5ms delay 10.00000% loss) ... (10.00Mbit 5ms delay 10.00000% loss)
*** Waiting for switches to connect
s3
Host h1 has IP address 10.0.0.1 and MAC adress de:84:cd:d4:36:06
Host h2 has IP address 10.0.0.2 and MAC adress f6:ee:19:30:7b:20
*** Running CLI
*** Starting CLI:
```

Figure 10: Запуск скрипта lab_iperf3.py



```
GNU nano 4.8 Makefile
all: iperf_result.json plot

iperf_result.json:
    sudo python lab_iperf3.py

plot: iperf_result.json
    plot_iperf.sh iperf_result.json

clean:
    -rm -f *.json *.csv
    rm -rf results
```

Figure 11: Создание Makefile

```
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ nano Makefile
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make clean
rm -f *.json *.csv
rm -rf results
mininet@mininet-vm:~/work/lab_iperf3/iperf3$ make
sudo python lab_iperf3.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
(100.00Mbit 75ms delay) (100.00Mbit 75ms delay) (100.00Mbit 75ms delay)
*** Starting network
*** Configuring hosts
h1 (cfs -l/1000000us) h2 (cfs -l/1000000us)
*** Starting controller
c0
*** Starting 1 switches
s3 (100.00Mbit 75ms delay) (100.00Mbit 75ms delay) ... (100.00Mbit 75ms
delay)
*** Waiting for switches to connect
s3
*** Starting network
*** Traffic generation
*** h2 : ('iperf3 -s -D -1',)
*** h1 : ('iperf3 -c', '10.0.0.2', '-J > iperf_result.json')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s3
*** Stopping 2 hosts
h1 h2
*** Done
plot_iperf.sh iperf_result.json
mininet@mininet-vm:~/work/lab_iperf3/iperf3$
```

Figure 12: Проверка работы Makefile

В результате выполнения данной лабораторной работы я познакомился с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получил навыки проведения воспроизводимого эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

1. Mininet [Электронный ресурс]. Mininet Project Contributors. URL: <http://mininet.org/> (дата обращения: 17.11.2024).
2. IPerff [Электронный ресурс]. URL: <https://iperf.fr/>.